

Paul Jialiang Wu, PhD

Forward Deployed AI Engineer — Enterprise Agentic Systems · Grounded RAG · LLMOps Gates

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SUMMARY

I embed with customers and take agentic systems from prototype to production-grade, and I ship the evidence with the system: golden datasets as permanent CI gates, every decision citing the clause it relied on, escalation to a human gated at exactly 1.0, and refusal as a first-class outcome. Then I codify what worked into tooling the next engineer installs in one command.

WHY CLOUDERA — FORWARD DEPLOYED AI ENGINEER (SENIOR/PRINCIPAL)

- **Enterprise agentic app taken from brief to gated evidence — HITL, retries, audit trail:** Triages purchase requests against six fragmented policy documents — role-based approval limits, vendor risk tiers, PII addenda, a v3→v4 amendment that supersedes prior rules. Durable-execution StateGraph from the primitive up (~170 lines): checkpoint per super-step, human-in-the-loop interrupt/resume, per-node retry, bounded self-correction on ungrounded decisions. 59 labeled cases across 11 tag families incl. prompt injection. decision_accuracy 1.0 · escalation_recall 1.0 (gated at exactly 1.0) · citation_validity 1.0 · trace_completeness 1.0, a permanent CI gate. Its first golden run FAILED (0.889) and exposed an unimplemented policy rule — a golden set that cannot fail you is not measuring anything. Live playground on a real API. (github.com/wjlgatech/FDE-os/tree/main/take-home/enterprise-agentic-triage · enterprise-triage-brief.vercel.app)
- **Grounded RAG over real filings where refusal is a first-class outcome:** Natural-language questions over real SEC filings as PDFs (Tesla 10-K + two 10-Qs, Apple 10-K) — SHA256-pinned, extracted to page-anchored JSON. Verbatim quotes with page anchors, deterministic arithmetic instead of model arithmetic, refusal as an explicit path. 26 hand-verified golden cases, four gates all 1.0: answer_accuracy, refusal_recall (unanswerable questions MUST refuse), refusal_precision, citation_validity (every quote re-verified verbatim). Cross-filing reconciliation: Q1+Q2 = 6M YTD exact on revenue and net income across two independent documents. Shipped behind an edge-middleware access gate; CI fully offline. (github.com/wjlgatech/FDE-os/tree/main/take-home/sec-filing-intelligence)
- **The eval framework that decided production model selection:** Primary technical interface on an executive-level enterprise AI deployment (Accenture OpenAI-DSI). Built the multi-model evaluation framework comparing 6+ VLMs (Gemini-class, NVIDIA VSS, Cosmos-Predict2) on accuracy, latency under load, edge-case failure modes and integration cost — the basis for production model selection. Shipped a confidence-calibrated selective-verification pipeline routing only high-uncertainty outputs to human review: ~90% annotation cost reduction. (Accenture, Physical AI Team (2024–present))
- **Codified the delivery patterns into installable tooling other engineers run:** FDE-os turns engagement patterns into reusable product: eleven skills + two composition workflows, exposed as a dependency-free MCP server (eleven callable tools), installable as a one-command Claude Code plugin and running unchanged on Codex and Hermes from one skills tree. Includes an offline RAG/agent eval harness (precision@k, recall@k, MRR, grounding proxy, citation coverage, CI gate), a self-improving eval-loop primitive that reverts regressions, and an enterprise doc parser with a parse-quality NO-GO gate. Live no-login webapp plus a shared-key model proxy, running at ~\$0. (github.com/wjlgatech/FDE-os · wjlgatech.github.io/FDE-os)
- **Multi-agent business-process automation in production for enterprise clients:** Event-sourced state management for concurrent-agent consistency, idempotent retry across distributed agents, and a full audit trail — live autonomous workflows with zero-downtime replanning. The core agentic failure mode is that agents over

partially observable state must replan without losing context; the event-sourced consistency model solves it and doubles as the tracing/observability surface. (Accenture, WorkflowX / Agenticom (2024–present))

ALSO SHIPPED

- **Compiled a data platform's whole curriculum into a staffing GO/NO-GO** — github.com/wjlgatech/FDE-os/tree/main/snowflake-os
- **Self-improving harness with an independent referee** — github.com/wjlgatech/loop-engineering-anything
- **Spatial AI deployed to Fortune 500 field technicians** — Accenture, Physical AI Team

EXPERIENCE

Applied AI Scientist / Forward Deployed Engineer — Accenture, Physical AI Team · Jan 2024–Present

- Primary technical interface for an executive-level enterprise AI deployment (OpenAI-DSI engagement); recognized for a "no surprises" standard built on deep model understanding and contextual judgment at decision points.
- Built the multi-model evaluation framework comparing 6+ VLMs (Gemini-class, NVIDIA VSS, Cosmos-Predict2) on accuracy, latency under load, edge-case failure modes and integration cost — the basis for production model selection.
- Shipped a confidence-calibrated selective-verification pipeline routing only high-uncertainty outputs to human review: ~90% annotation cost reduction.
- Technical lead on 4 simultaneous production systems: synthetic data generation, zero-shot safety detection, adaptive evaluation under distribution shift, vision-language reasoning over live sensor streams.
- Led technical discovery and white-glove deployment: translated ambiguous customer requirements into shipped agentic workflows, and co-built with customer engineering teams to instill rigorous development practices and drive end-user adoption.
- Designed and shipped a multi-agent business-process automation engine for enterprise clients (WorkflowX / Agenticom): event-sourced state for concurrent-agent consistency, idempotent retry across distributed agents, full audit trail, zero-downtime replanning.

Principal Data Scientist — Genentech · Jun 2021–Dec 2022

- Production ML for biomedical research; 3 open-source PyPI packages for multimodal learning; 85% reduction in unproductive process time.

Principal Data Scientist — Galvanize Inc. · Sep 2019–Jun 2021

- Technical lead across 13+ end-to-end ML systems for Fortune 500 clients; 5-star-rated delivery.

PUBLICATIONS

- **Self-Calibrating World Models (NeurIPS 2026, under review)** — Online Bayesian calibration for sim-to-real transfer; 40% transfer-error reduction vs domain randomization.
- **Physical AI: The Next Frontier in AI and Robotics** — Preprints.org, Apr 2026. DOI: [10.20944/preprints202604.0549.v1](https://doi.org/10.20944/preprints202604.0549.v1).
- **Failure Benchmarking of NVIDIA's VSS Tool: Insights from Vision-Language Evaluation** *(working paper · first author · in preparation)* — Wu, P.J., Shah, A., Hosseini, P., Shahab, S. — Center for Advanced AI, Accenture. A five-mode VLM failure taxonomy made executable over deterministic synthetic stimuli, with the

Temporal Grounding Score defined as data. Baseline VSS fails 5/5 modes; two frontier models pass 5/5 and the smallest reproduces 3/5, so the failures track capability, not only late-fusion architecture. No venue selected.

- **Adapting the BARE Framework for Synthetic Data Generation in Vision-Language Models** *(working paper · first author · in preparation)* — Wu, P.J., Shah, A., Hosseini, P., Shahab, S. — Center for Advanced AI, Accenture. Base-model diversity + instruction-tuned correctness as a twin-gated loop. Mode collapse reproduced (0.23 vs 0.62); base-model hallucination did not. Our own first negative was retracted as underpowered (0/18 $\Rightarrow$ CI [0, 16.7%]) and re-resolved at n=66 and n=72 as a genuine one. No venue selected.
- **Eval gates as training-signal factories (RewardForge)** *(working paper · unpublished, not submitted)* — Every honest eval gate both certifies quality and labels preference data. 162 pairs from certified gates, zero human labels $\rightarrow$ DPO from scratch $\rightarrow$ LoRA $\rightarrow$ held-out hallucination 0.398 $\rightarrow$ 0.287; five seeds against a pre-committed criterion, treatment $+0.098 \pm 0.028$ vs shuffled-label control -0.046 ± 0.055 .
- **Evidence-gated benchmarking of agentic tool use (MCP-Arena)** *(working paper · unpublished, not submitted)* — 12 tasks $\times$ 6 failure-mode categories over real MCP servers, asserting the tool-call trace rather than the prose, so a fluent answer with no tool calls scores zero by construction; ships a hallucinating mock that must score 0 in CI.
- **Predicting a frontier lab's post-training stack, scored rather than asserted** *(working paper · 12 predictions registered, Brier UNSCORED)* — A ten-stage reconstruction of a frontier lab's training stack read from its publications, with 12 falsifiable claims registered 2026-08-13 and resolving by 2026-12-31 against the eventual technical report. 0 of 12 resolved, so the Brier score is reported UNSCORED rather than estimated.
- **The zero that wasn't evidence — power discipline for gates that report negatives** *(methods note · unpublished, not submitted)* — An audit of our own false negative and the rule it produced: wherever a gate reports a negative it must state the n and the smallest effect that n could have resolved — with a runnable tool (Wilson intervals, rule of three, minimum detectable difference), because a rule without one is a slogan.

TECHNICAL SKILLS

Agentic Systems: Multi-agent orchestration (ReAct, self-reflection, hierarchical delegation), durable-execution state graphs (checkpoint/resume, HITL interrupt, retry policy, bounded self-correction), event-sourced agent state, MCP servers and clients, LangGraph/CrewAI/ADK-class patterns

Retrieval, RAG & Data: RAG over structured + unstructured enterprise data, semantic search, embeddings and vector databases, page-anchored citation verification, deterministic arithmetic over retrieved facts, DOCX/XLSX/PDF/web/transcript ingestion with parse-quality gates, provenance-pinned corpora

LLMOps / MLOps: Independent-referee eval pipelines, golden datasets as permanent CI gates, grounding/hallucination and citation-validity metrics, prompt-injection probes, granular tracing and audit trails, confidence calibration, distribution-shift detection, cost/latency/token budgets, regression rollback, model-provider seams for portable swaps

Cloud & Infrastructure: GCP (hands-on ML + Gemini evaluation), Azure ML (production), AWS; NVIDIA AI stack (NIM, Cosmos, VSS); Docker, Kubernetes, CI/CD, Vercel edge middleware and RBAC gating, FastAPI, React, SQLite/Postgres

Forward-Deployed Delivery: Technical discovery, executive stakeholder alignment, white-glove deployment, prototype $\rightarrow$ production hardening, customer engineering enablement, reference architectures and starter kits, field-pattern $\rightarrow$ reusable module $\rightarrow$ product feedback loop

EDUCATION

Yale University — National BioMed Fellow, Computational Immunology · **Georgia Tech** — PhD, Bioinformatics · **University of South Carolina** — MS, Mathematics & CS · **Sun Yat-Sen University** — BS, Applied Mathematics

THE EDGE I'D BE LEARNING, NOT TEACHING

I have not run the Cloudera platform itself, and my big-data-infrastructure depth is the real edge: I have built RAG and agentic pipelines over enterprise documents and live sensor streams, but I have not owned a Spark/Iceberg/NiFi data platform in production, and my inference work is model selection, cost/latency budgeting and serving seams rather than GPU kernel-level optimization. That is the part of this role I would be ramping on in the first weeks, not teaching.